

Grade 7 Accelerated
Unit 1
Lesson 10 Homework

Name _____
Date _____
Period _____

Quinn solved the inequality $2.5 \geq \frac{c-3}{4}$, but made an error. Her work is shown.

Quinn's work

$$\begin{aligned} 2.5 &\geq \frac{c-3}{4} \\ 2.5 + 3 &\geq \frac{c-3+3}{4} \\ 5.5 &\geq \frac{c}{4} \\ 22 &\geq c \end{aligned}$$

1. What error did Quinn make?
2. Write a brief note to Quinn explaining how to avoid making the same mistake in the future. Be sure to include what she did correctly in your note.
3. Solve Quinn's inequality correctly and represent the solution on a number line.

For questions 4-7, solve each inequality and represent the solution on a number line.

4. $-\frac{b}{4} + 5 > \frac{1}{2}$ 

5. $8.8 \geq 0.8(-17.3 + 2.9x)$ 

6. $1.5m - 10 < \frac{1}{-4}$



7. $-\frac{2}{3}(-3 - 12a) \leq -6$



Use the following scenario for questions 8-10.

Mary is buying donuts and juice for her son Joey's 3rd birthday party at school. She has at most \$25 to spend. She bought 10 donuts for \$1.25 each and each case of juice costs \$3.49. Mary wrote the inequality $10(1.25) + 3.49c \leq 25$, where c represents the number of cases of juice she can purchase.

8. Why did Mary use the inequality symbol $\leq$ instead of $<$, $>$, or $\geq$?

9. Solve the inequality.

10. Will Mary be able to purchase 4 cases of juice? Justify your reasoning mathematically.